

Disaster Relief Logistics

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Abstract

This project analyzed the complex interaction between social media and disaster relief efforts following the devastating 2023 Kahramanmaraş earthquake in Turkey. We investigate how social media was used both on an individual and organizational level during this crisis. First, user-generated content is examined for the development of a topic modeling approach to classify the primary needs of disaster relief supplies. In the second case, we will review the response of different multinational brands to the earthquake on the social media platform by examining the engagement level and types of relief. We will also highlight how social media platforms play a key role in connecting people and showing support through active participation and bringing out their needs that are required to be recognized in the event of a disaster, though sometimes there are barriers and challenges.

Keywords: Brand Engagement, Social Media

1 Introduction

The 2023 Kahramanmaraş earthquake in Turkey was a devastating event that underscored the critical importance of effective disaster relief logistics. In the aftermath of such crises, social media platforms play a crucial role in facilitating communication, sharing information, and mobilizing resources (Eriksson, 2018). While these platforms connect people to seek help and coordinate relief activities, businesses also spring up to respond to such situations. Our project takes a critical explanation of social media during disaster relief logistics, especially how multinational brands leveraged social media to communicate their response towards, or engage in, afflicted communities' benefit during the Kahramanmaraş earthquake.

Basically, we focus on two major areas of how social media were used during the earthquake:

Disaster Relief Needs Identification:

Brand Response and Engagement: We analyzed how each of these multinational brands responded to the earthquake on social media platforms through their communication strategy, likes, shares, comments obtained about the earthquake-related posts so the total engagement was examined.

Our findings add to the improved insight into the emerging role of social media in disaster response by underlining key points for improvement for future responses. Apart from that, our findings introduce a major focus on strategic communication during disasters by global brands and especially its impact on resource mobilization and support in disaster contexts.

2 Methodology

2.1 Brand Response and Engagement

The initial research plan had focused on tweets that related to the 2020 Izmir earthquake since the dataset at hand was up to 2021. However, after manually checking the dataset provided by our instructor we found that none of the selected brands tweeted about this event.

Thus, we set our focus on the 2023 Kahramanmaraş earthquake. Our instructor then provided us with a dataset comprising tweets made in February 2023 with variables for Tweet Text, User Information, Timestamps, and Engagement Metrics-likes, retweets, and comments.

To make sure that our analysis is right and differentiate the brand communication related to the earthquake, we had an important data preparation step. We manually classified each tweet as "related to the earthquake" (coded as 1) or "not related to the earthquake" (coded as 0). So we added a new column "Deprem" to our data set which helped us analyze brand activities and engagements related to the earthquake specifically.

We chose a sample of prominent multinational brands with a presence in Turkey, focusing on those with significant social media presence. Our selection was based on the brands' visibility, relevance to the Turkish market, and their general engagement with their audiences on Twitter. The brands included in our sample were:

- Vodafone Türkiye
- TURKCELL
- Siemens Türkiye
- McDonald's Türkiye
- CarrefourSA
- Watsons Türkiye
- UPS Türkiye
- Burger King Türkiye
- Flormar Türkiye
- Samsung Türkiye
- Marks and Spencer TR
- AXA Sigorta

- Algida Türkiye
- Yves Rocher Türkiye
- Hyundai Türkiye
- Huawei Türkiye
- BMW Türkiye
- Netflix Türkiye
- Microsoft Türkiye
- KFC Türkiye
- Gillette Türkiye
- H&M Türkiye
- Toyota Türkiye
- DHL Express Türkiye
- Pepsi Türkiye
- HSBC Türkiye
- Coca-Cola Türkiye

This diverse range of brands represents various sectors, including telecommunications, fast food, retail, and financial services. Their presence in Turkey and active social media engagement made them valuable subjects for examining brand responses.

Before going into details on brand engagement, we need to present a broader picture of the social media activity related to the Kahramanmaraş earthquake. To understand how widespread earthquake-related content is, we analyzed the frequency of tweets related to the earthquake compared to non-related tweets.

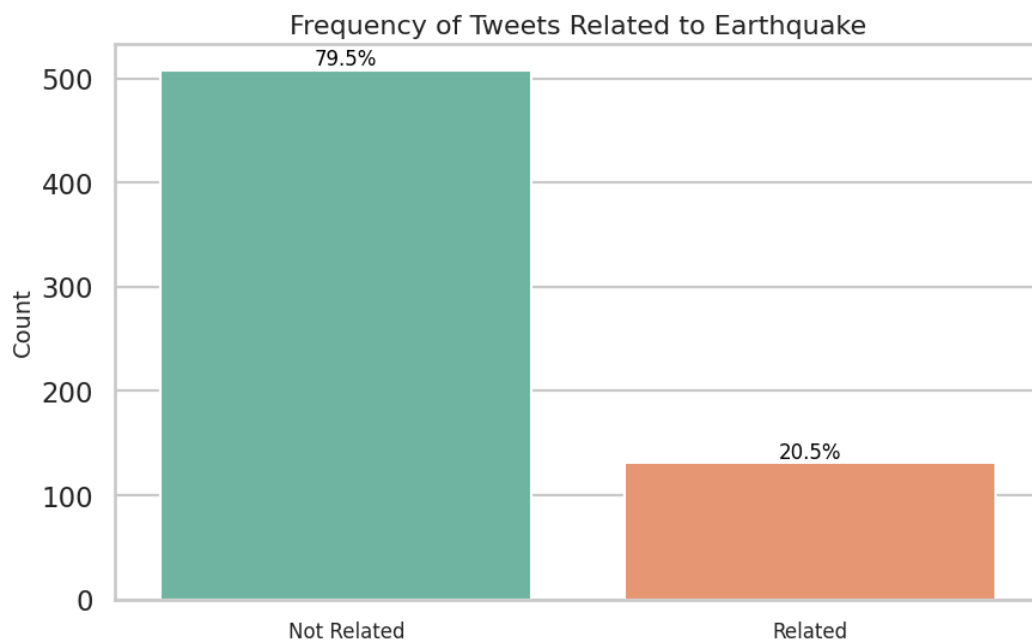


Figure 1: Frequency of Tweets Related to Earthquake

This graph shows a surprising imbalance: the overwhelming majority (79.5%) of all tweets in our data set were not earthquake-related. A first implication of this observation is that, while brands were active on social media during the disaster, much of their communication did not address the earthquake or related relief efforts directly. This

further underlines how brand communication analysis should be done more critically in terms of the quantity and quality of the content in disasters.

Then the number of tweets related to the earthquake that each brand posted was calculated and graphed.

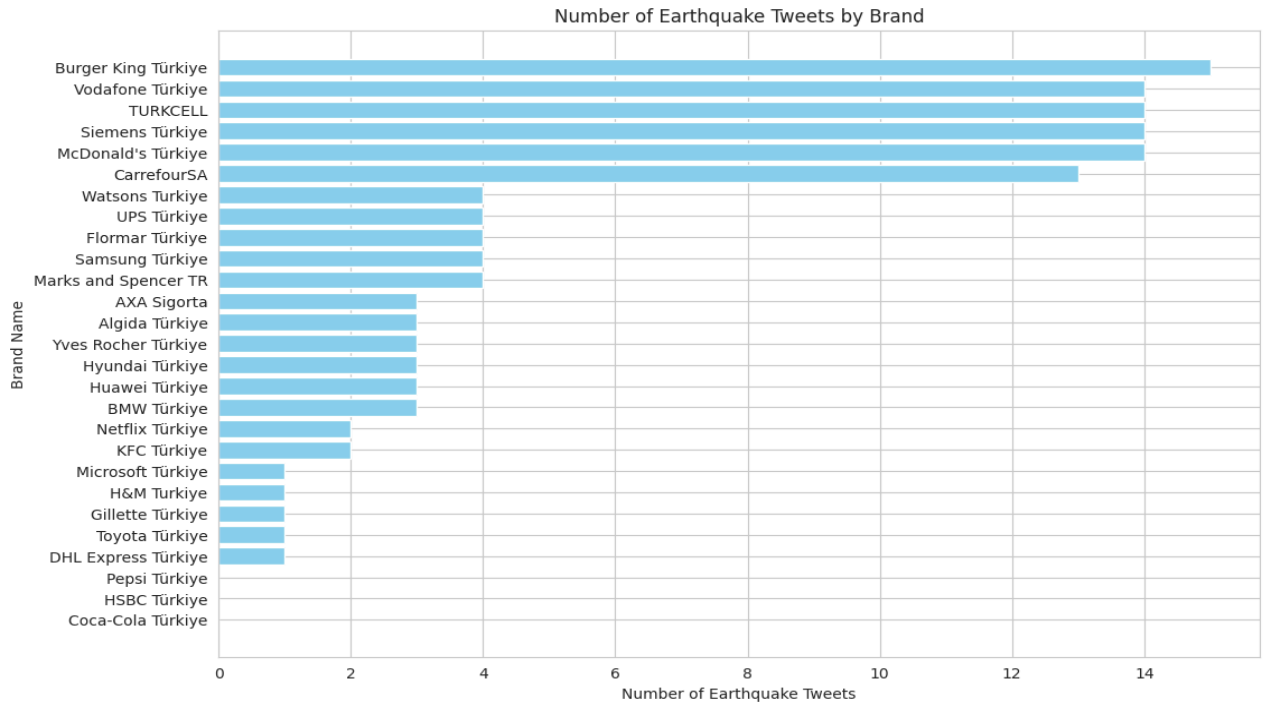


Figure 2: Number of Earthquake Tweets by Brand

This visualization provided a direct measure of the brand's level of activity regarding the disaster. The second and third of the most tweeted brands are Vodafone and Turkcell, which are major GSM operators and in the first month of the earthquake, communication was crucial for the coordination of the relief efforts and communicating with the loved ones. Also, Burger King is first on the list because it played a significant role in providing meals and essential support to the affected areas. Its high level of activity on social media reflects its commitment to disaster relief, with frequent updates on its efforts to assist those impacted by the earthquake.

To further understand the overall impact of the earthquake on social media, the daily count of earthquake-related tweets were graphed.

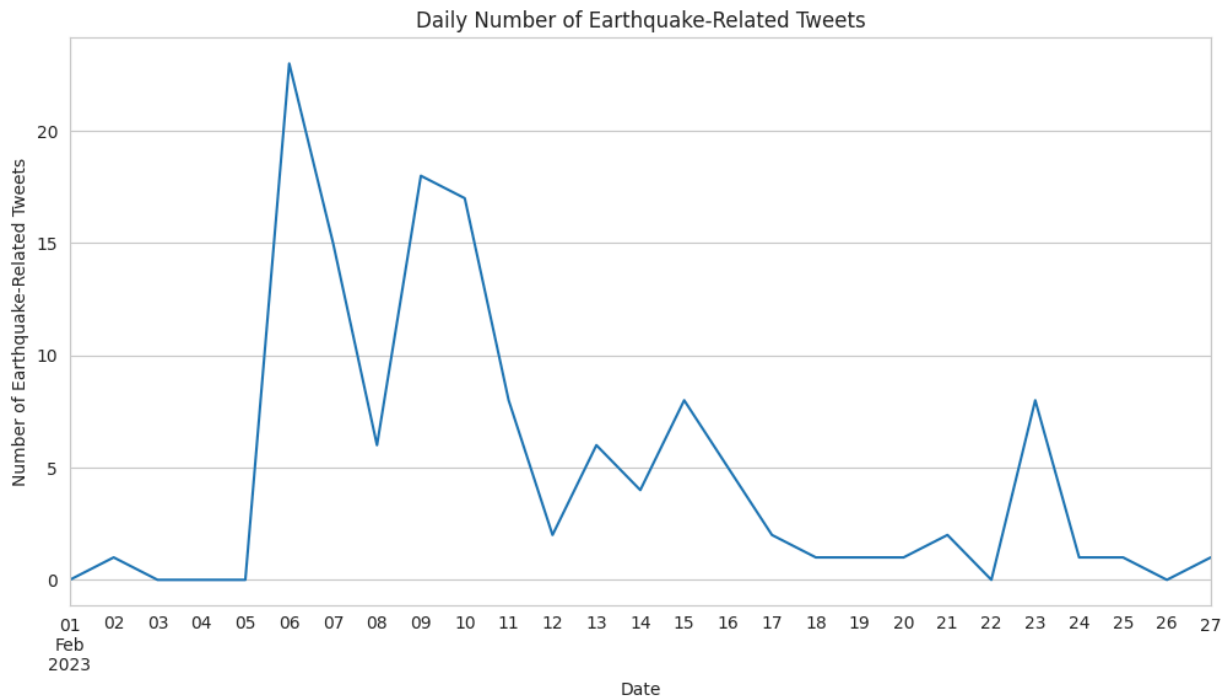


Figure 3: Daily Number of Earthquake-Related Tweets

This figure shows that interactions over earthquake-related content peaked on February 6, on the exact date of the earthquake. The result may be seen as showing the immediate and overwhelming reaction towards the event as brands sought to share, show solidarity, and rally for support in their own way. In the days that followed, engagement slowly decreased, pointing out that the initial intense focus on the event gave way to more sustained efforts related to long-term relief and recovery. This trend highlights the typical pattern observed in disaster response, where public interest is initially high but can diminish as media attention shifts and the urgent phase of the crisis passes.

To gain deeper insights into the relationship between engagement metrics and specific tweet characteristics, correlation analysis using the Seaborn library was conducted. The result revealed a slightly stronger correlation of earthquake-related tweets with engagement metrics when compared to unrelated tweets. This may suggest that the content about the disaster resonated more powerfully with audiences and thus demanded more interaction.

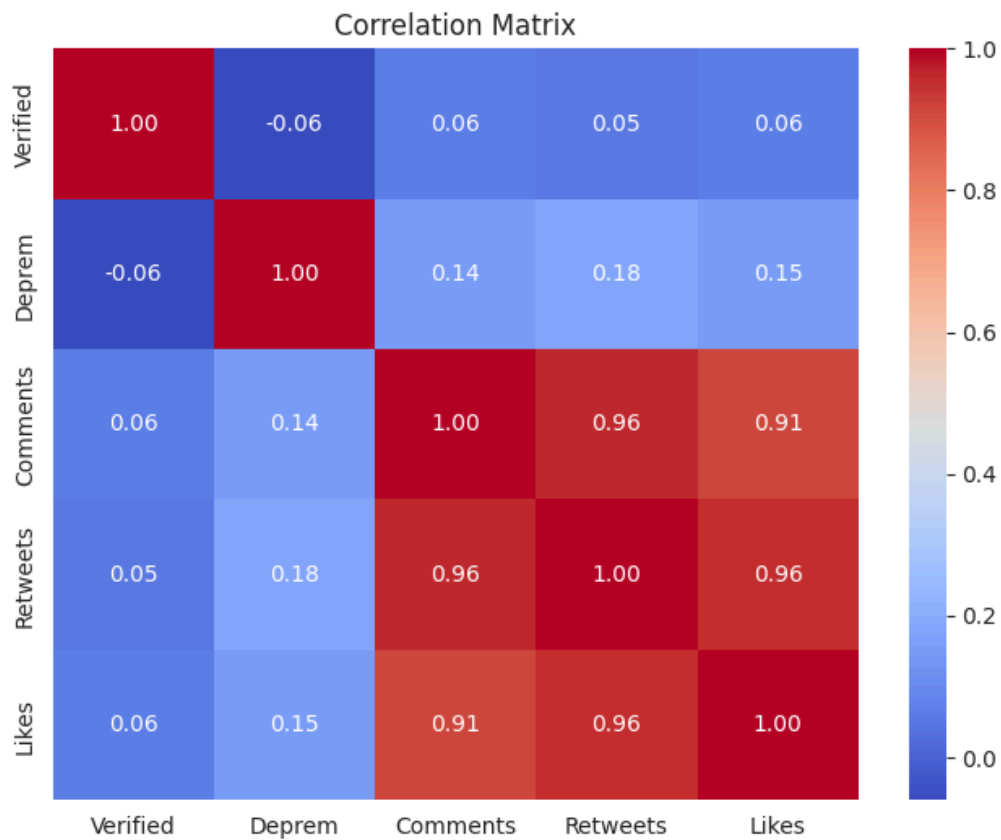


Figure 4: Correlation Matrix

To further investigate this relationship, we performed an Ordinary Least Squares (OLS) regression analysis using the statsmodels.api library. In our regression model, we included two independent variables: 'Deprem' (a binary variable indicating whether a tweet was related to the earthquake) and 'Verified' (a binary variable indicating whether the account was verified). The dependent variable was 'Engagement,' a composite metric derived from the number of likes, retweets, and comments.

The results showed that earthquake-related content ('Deprem') was actually a predictor of engagement, showing a positive coefficient and a statistically significant p-value. This suggests that the topic of the earthquake was more important for engagement than the account verification status; hence, content related to disasters could be much more engaging for an audience.

Finally, to provide a visual representation of brand engagement with the earthquake, we created a bar chart showing the total engagement for each brand, again the engagement is the total number of retweets, likes and comments.

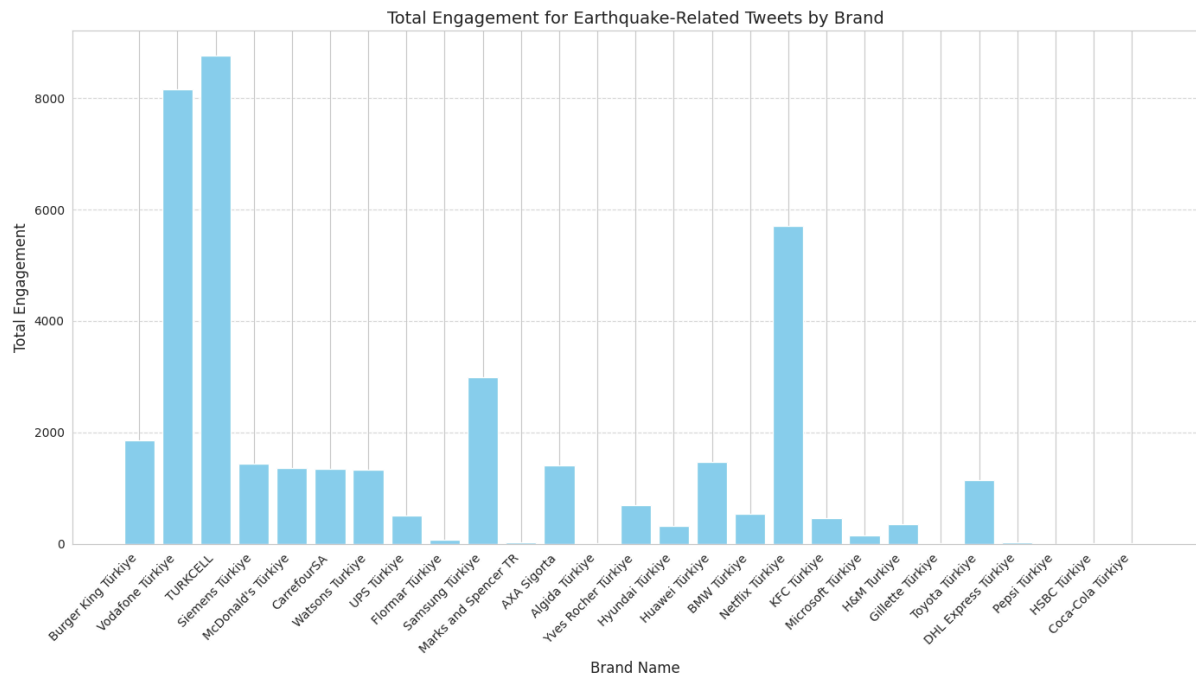


Figure 5: Total Engagement for Earthquake-Related Tweets by Brand

The figure shows that brands with more followers within the social network, such as Vodafone and Turkcell, received the highest engagement scores. This result is as expected since both were major GSM operators and people needed communication during the time of the earthquake the most. Also it can be concluded that high engagement does not always mean a high number of tweets. A brand such as Burger King created many tweets but didn't get as high of an engagement score.

Lastly by manually examining the data, we saw that most multinational brands played a part in earthquake relief efforts. Each contributed significantly but did so in particular ways related to the industry type in which they were involved. Financial contributions were one of the more common ways: Hyundai, KFC, and Netflix all donated. CarrefourSA and Burger King supplied the essentials: food and generators. Logistics companies UPS and DHL provided transportation support. Communications companies Turkcell and Vodafone ensured people could be reached with free calls and access to the Internet. Some brands, like Huawei, did no more than release a message of solidarity, with no real assistance. On the whole, it emerged that the providers of essential goods and express services were the most aggressive among all in their relief operations.

3 Discussion and Conclusion

In conclusion, this analysis brought out in light how social media plays a rather crucial role in disaster management: to spread information, coordinate aid, or provide support at the community level. By examining brand communication on social media during crises, we unlock profound insight into the effectiveness of various methods and the drivers of online activities in such times. Further studies could investigate in more detail how the type of content, the demographic distribution of the audience targeted, and specific functionalities of each platform influence disaster relief communication to extend our

knowledge concerning what can be done with social media in a crisis situation.

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